

SRM Institute of Science and Technology

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College of Engineering and Technology

School of Electrical and Electronics

Department of Electronics and Communication Engineering

MINI PROJECT REPORT

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Project Title : RFID-BASED ATTENDANCE SYSTEM

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Particulars	Max. Marks	Marks Obtained
		Name:
		Register No:
Review 1 and 2	05	
Demo verification & viva	03	
Project Report	02	
Total	10	

Date :

Staff Name :

Signature :

RFID-BASED ATTENDANCE SYSTEM

OBJECTIVE:

The objective of this project is to design and implement an RFID-based attendance system using the NODEMCU ESP8266 microcontroller. The system aims to streamline attendance tracking by scanning RFID tags, which automatically logs attendance data, including the date, time, and registration number, into a Google Sheet.

ABSTRACT:

This project presents an RFID-based attendance system using the NODEMCU ESP8266 microcontroller, aimed at automating and simplifying the process of attendance tracking. The system employs an RC522 RFID reader to scan RFID tags, each uniquely associated with an individual. When a tag is scanned, the system records the attendance data, including the date, time, and registration number, which is then uploaded to a Google Sheet via WiFi. A 16x2 I2C LCD display confirms the successful logging of each entry, while a buzzer provides immediate auditory feedback upon scanning. This automated system eliminates the need for manual attendance entry, enhancing data accuracy and enabling efficient, real-time monitoring. The project demonstrates a practical, scalable solution for attendance management, with applications in educational institutions, offices, and event settings.

INTRODUCTION:

In this project, we address the development of an RFID-based attendance system, which provides a modernized approach to tracking and managing attendance in real-time. Traditional attendance methods, such as manual sign-ins or roll calls, are often time-consuming, prone to errors, and can be challenging to manage efficiently, especially in environments with a large number of attendees. This project leverages RFID technology and the internet-enabled NODEMCU ESP8266 microcontroller to streamline the process. When an RFID tag is scanned, attendance information—including date, time, and registration number—is automatically stored in a Google Sheet. This real-time, automated approach reduces human error, enhances accuracy, and simplifies data management, making the system highly effective for applications in schools, universities, offices, and event management. By implementing this solution, users can effortlessly record and access attendance data, promoting a seamless and efficient attendance management process.

HARDWARE/SOFTWARE REQUIREMENTS:

Hardware details

1. NODEMCU ESP8266 Microcontroller: - A WiFi-enabled microcontroller that serves as the central processing unit for the project. It connects to the internet to upload attendance data to Google Sheets, making the data accessible in real time.

2. RC522 RFID Reader:

- This module is used to read RFID tags by scanning them when they are brought close to the reader. The RFID reader communicates with the NODEMCU to transfer the unique ID of each tag, which is used to identify attendees.

3. RFID Tags:

- Passive RFID tags are uniquely associated with each individual. When scanned by the RC522 reader, the unique ID of the tag is detected and used to log attendance.

4. 16x2 I2C LCD Display:

- The LCD displays messages to the user, such as "Attendance Recorded" or error messages, providing real-time feedback for each scan. The I2C interface simplifies connections with fewer pins required, making it easy to connect to the NODEMCU.

5. Buzzer:

- The buzzer provides an audible signal whenever an RFID tag is successfully scanned. This feature enhances user feedback by confirming the successful recording of attendance.

6. Breadboard and Jumper Wires:

- The breadboard is used to set up connections between the components without soldering, and jumper wires facilitate these connections. This setup is ideal for prototyping and testing the circuit before final implementation.

Software details

1. Arduino IDE:

- The Arduino IDE is used to write, compile, and upload code to the NODEMCU ESP8266 microcontroller. This open-source platform supports various libraries needed for handling the RFID reader, LCD, WiFi connection, and Google Sheets API integration.

- Key libraries include:

- ESP8266WiFi.h: For connecting the NODEMCU to WiFi.

- MFRC522.h: To interface with the RC522 RFID reader.

- Wire.h and LiquidCrystal_I2C.h: For controlling the 16x2 I2C LCD display.

2. Google Sheets API:

- Google Sheets API allows the system to upload attendance records directly to a Google Sheet. By using Google Apps Script or a suitable API call, the project enables real-time data logging with date, time, and registration number.

- This requires setting up a Google Sheets project and enabling the Google Sheets API to communicate with the ESP8266.

3. Google Apps Script :

- This scripting platform can be used to create a web app within Google Sheets to handle data submissions from the ESP8266. The script will parse incoming data and log it into the sheet with automated date and time stamps.

4. WiFi Connection: - The NODEMCU ESP8266 connects to the local WiFi network, enabling data transfer to Google Sheets. The project's code includes WiFi configuration settings to connect securely to the specified network.

5. JSON Data Format:

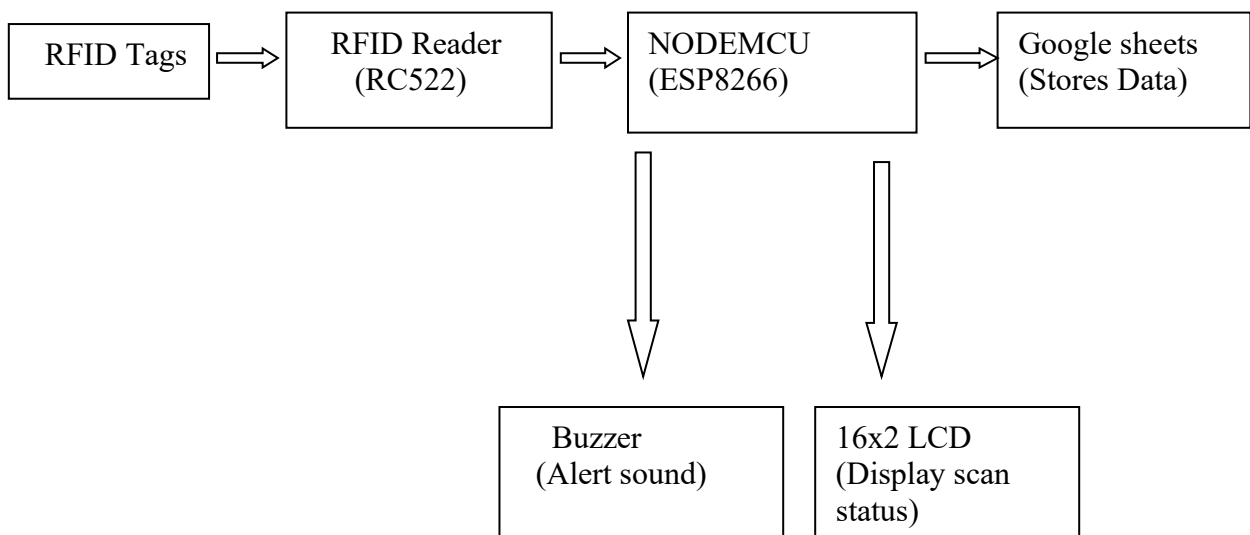
- Data from the RFID reader is sent to Google Sheets in JSON format, which organizes the registration number, date, and time in a structured way for easy parsing and storage.

6. Libraries Installation and Configuration:

- All necessary libraries (such as ESP8266WiFi, MFRC522, and LiquidCrystal_I2C) must be installed in the Arduino IDE and configured in the code. Initial setup includes defining variables for WiFi credentials, Google Sheets URL, and RFID reader settings.

CONCEPTS/WORKING PRINCIPLE

Block diagram:



RFID Tag: Scanned by the RFID Reader.

RFID Reader: Detects the RFID tags when they are placed close to the reader.

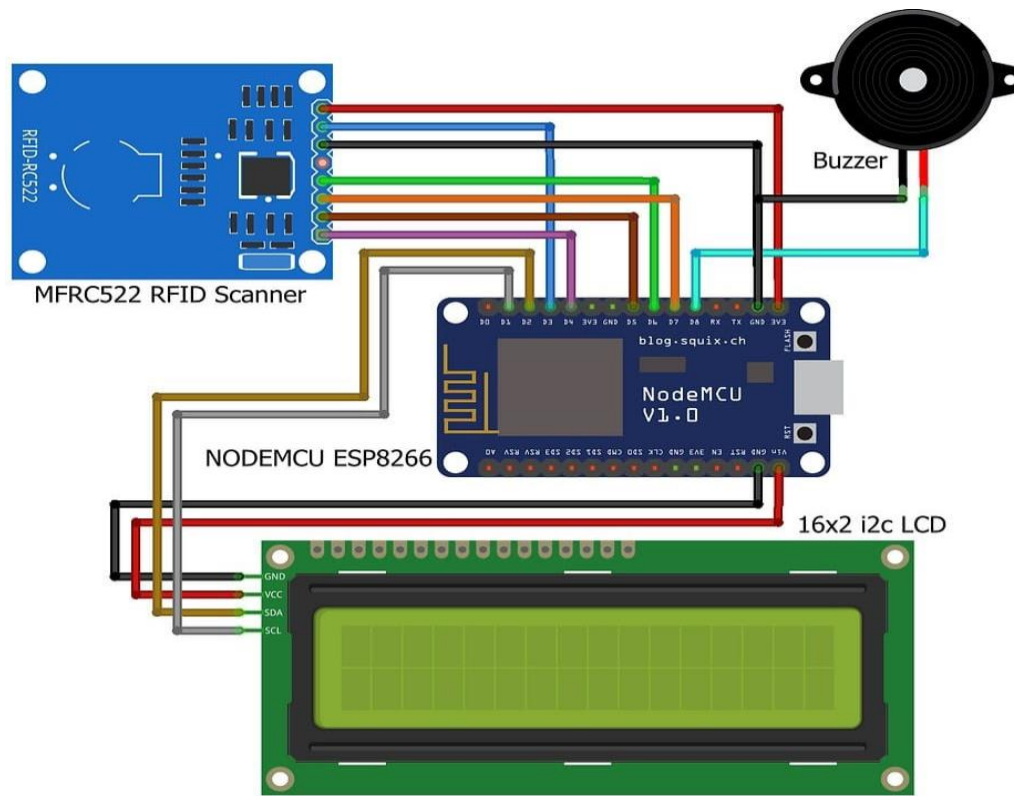
Microcontroller: Receives the RFID data, processes it, and sends the attendance information to Google Sheets.

LCD Display: Shows messages to inform the user about the status (like successful attendance recording).

Buzzer: Emits a sound to provide feedback to the user upon successful scanning of the RFID tag.

Google Sheets: Acts as a cloud database to store attendance records including date, time, and registration number.

Circuit Diagram:



Components and Connections

1. NodeMCU ESP8266: This is the microcontroller that coordinates the reading of RFID tags, displays messages on the LCD, and controls the buzzer.

2. MFRC522 RFID Reader: The RFID reader module scans the RFID tags. Here's how it connects to the NodeMCU:

- ****SDA**** connects to ****D2**** on the NodeMCU.
- ****SCK**** connects to ****D5**** on the NodeMCU.
- ****MOSI**** connects to ****D7**** on the NodeMCU.
- ****MISO**** connects to ****D6**** on the NodeMCU.
- ****IRQ**** is not connected in this setup.
- ****GND**** connects to ****GND**** on the NodeMCU.
- ****RST**** connects to ****D1**** on the NodeMCU.
- ****3.3V**** connects to the ****3.3V**** pin on the NodeMCU.

3. 16x2 I2C LCD Display: The LCD display is used to show messages, such as "Ready for Scan" or "Attendance Marked".

- ****GND**** connects to ****GND**** on the NodeMCU.
- ****VCC**** connects to ****3.3V**** on the NodeMCU.
- ****SDA**** connects to ****D3**** on the NodeMCU.
- ****SCL**** connects to ****D4**** on the NodeMCU.

4. Buzzer: The buzzer provides audible feedback when an RFID tag is successfully scanned.

- The positive terminal of the buzzer connects to ****D0**** on the NodeMCU.
- The negative terminal connects to ****GND**** on the NodeMCU.

5. Power and Ground Connections

- Both the RFID reader and LCD share the **3.3V** and **GND** lines with the NodeMCU, ensuring they receive the correct voltage.

APPROACH/METHODOLOGY/PROGRAMS:

Algorithm for RFID-Based Attendance System with Google Sheets Integration

1. Setup Phase:

- Initialize the **NodeMCU ESP8266** to connect to a **Wi-Fi network**.
- Initialize the **RC522 RFID reader** for tag scanning.
- Set up the **I2C LCD display** for output messages.
- Configure the **buzzer** for feedback on successful scans.

2. Main Program Loop:

- Step 1: Connect the NodeMCU to Wi-Fi.
 - If Wi-Fi connection is unsuccessful, retry until connected.
- Step 2: Display "Ready for Scan" on the LCD screen.
- Step 3: Wait for an RFID tag to be placed near the RC522 reader.

3. Tag Detection and Data Retrieval:

- Step 4: If a tag is detected, read the **unique ID** of the RFID tag.
- Step 5: Use the unique ID as the **registration number**.

4. Attendance Logging:

- Step 6: Retrieve the current **date and time** from the NodeMCU's real-time clock or the system's built-in clock.
- Step 7: Display "Attendance Marked" along with the registration number on the LCD.
- Step 8: Trigger the buzzer for a short beep to confirm attendance.

5. Data Storage in Google Sheets:

- Step 9: Format the **date, time, and registration number** into a data packet.
- Step 10: Send this data packet to Google Sheets using an HTTP POST request with the Google Sheets API.
 - Ensure the NodeMCU is authenticated with the appropriate API credentials for Google Sheets.
 - If the data transfer is successful, display "Attendance Recorded" on the LCD.
 - If unsuccessful, display "Error" and retry the upload.

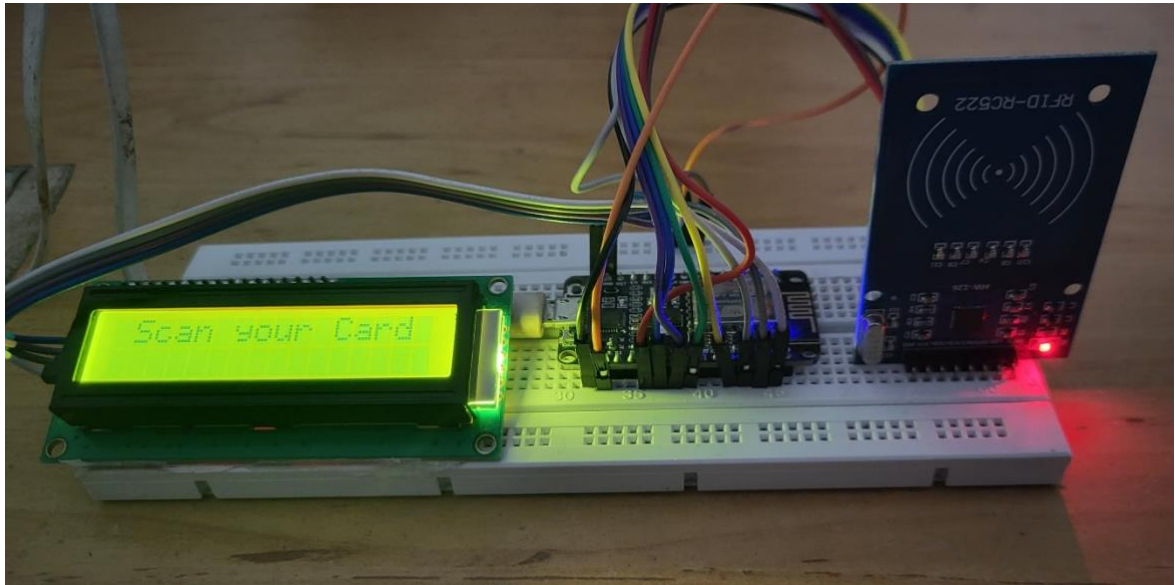
6. Repeat Process:

- Step 11: After recording, clear the LCD and reset to "Ready for Scan".

7. End of Program:

Await Next Scan: Return to the main loop to continuously await the next tag scan.

OUTPUT:



CONCLUSIONS:

Thus, the Rfid-based attendance system is constructed and system effectively automates attendance tracking by recording data in real-time to Google Sheets, enhancing efficiency and reducing manual errors.

REFERENCES:

https://www.youtube.com/redirect?event=video_description&redir_token=QUFFLUhqazM0d19CX24yM0JOTXFqSTNEaHdEaGIDclBTQXxBQ3Jtc0trQTdjeVBhYjBqTkEd05PTHliU0E4YjZjOWV4U0NuWk1UeVJUUVZreXQ1aHZVVUtTLU0yUzBfbWlib3pxSVRocmVELTd0d1JVT2labmJ6UllMci1JR2hqY1NLOHczRThEldUbXB3ZHFOhHWeEM2OA&q=https%3A%2F%2Fwww.viralsciencecreativity.com%2Fpost%2Fesp8266-rfid-attendance-system-with-google-sheets&v=B4yrRoj2hIM